introduced the concept of plant communities, and outlined how a group of species interact and develop in response to the same local conditions.

Plants and ecosystems

For many years, plant ecology and animal ecology were studied separately, but in the early 20th century a more connected perspective emerged. Important theories on plant communities and succession—the process by which an ecological community changes over time—were established during this time period. In 1926, Russian geochemist Vladimir Vernadsky introduced the idea of Earth's biosphere, the parts of its surface and atmosphere where all living organisms exist and interact.

Plants are sensitive barometers of change within an environment. The study of their anatomy, physiology, distribution, and abundance, as well as their interactions with other organisms and their response to environmental factors, such as soil conditions, hydrology, and pollution, can provide invaluable information about the entire ecosystem.

"That land is a community is the basic concept of ecology."

Aldo Leopold

American ecologist

See also: Climate and vegetation • Ecological succession • The biosphere • Endangered habitats • Deforestation